MANICOUAGAN, QUEBEC, PALEOMAGNETIC RESULTS

W. A. ROBERTSON

Geomagnetic Division, Dominion Observatory, Ottawa, Canada

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The remanent magnetization of dacitic material from Manicouagan, Quebec, was examined. Laboratory thermal studies support the view that viscous buildup in situ was small and secondary components insignificant. A pole position of 89° E., 57° N. is calculated from the mean direction of 6 sites. Comparison with other North American paleomagnetic poles points to a possible Triassic age for the formation of the material.

INTRODUCTION

The two semi-circular lakes of Manicouagan and Mouchalagan almost surround an area 35 miles in diameter 350 miles north by east from Quebec City (Fig. 1). A large hydro-electric dam system on the two rivers to the south has raised the levels in the two lakes and flooded the lower areas. Downfaulted blocks of Ordovician limestone, mapped before the flooding of the low-lying part, give a lower age limit to the formation of this feature.

The origin of this circular topographic feature has been explained by two conflicting hypotheses. The central anorthosite, originally interpreted as the intrusive body from which the dacitic lavas emanated (Rose 1955), is now recognized as part of the Precambrian basement. In a recent interpretation, Currie (1965) visualizes a series of initial phreatic eruptions followed by flows emanating from dykes and plugs distributed around the central mountain. The alternative hypothesis of meteorite impact was suggested during the search for craters in Canada (Beals, Innes and Rottenberg 1960). The anorthosite mountain is then regarded as part of the area of central uplift comparable with other supposed meteorite craters on the earth, the moon, and Mars, and the dacitic material is the product of shock melting and ejection from the region of impact (Dence 1965). The paleomagnetic study, initiated in September 1964, was an attempt to throw light on the mode of origin and age of the feature.

GEOLOGY AND SAMPLING

The circular feature is set in an area of Precambrian foliated gneisses, basic intrusive rocks, and metasediments of the Grenville (structural) Province. Most of the area within the twin lakes is covered by the dacitic material, which is in places more than 700 ft thick, and lies nearly horizontally on the basement. Mount de Babel, the prominent anorthositic central feature, rises to an elevation of more than 3000 ft. The dacite may be divided into a lower fine-grained and commonly inclusion-rich part, and an upper coarser-grained part in which inclusions are not so common. A K–Ar age of 225 ± 30 m.y. has been obtained from 'fine-grained andesite' closely associated with the dacitic rock, on the west shore of Manicouagan Lake (Wanless *et al.* 1966).

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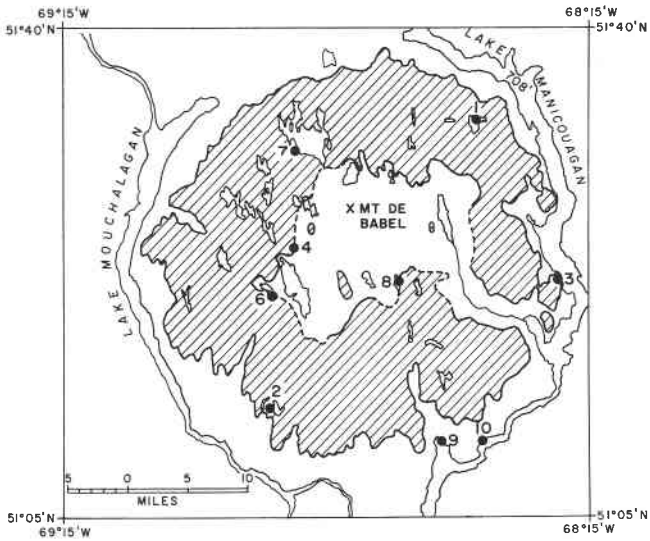


FIG. 1. Sketch map showing numbered sampling sites; lined pattern covers area of outcrop of dacitic material. The rock sampled at sites 9 and 10 is basement gneiss.

Two or more oriented samples were taken from each of 7 sites in the 'dacite', and also from 2 sites in the basement rock, using standard geological techniques. From these, two or more cylindrical specimens were cut using non-magnetic tools. Directions of magnetization were measured on an astatic magnetometer (Roy 1963*b*).

RESULTS

Directions before and after thermal cleaning are given in Fig. 2, and mean site directions in Table I. The directions from site 1 are random at $P = 0.05$ (Watson 1956) and have been excluded in the formation calculations. The pilot runs (Table II) showed very small direction changes with heating up to 350 °C (Fig. 3), using either sites or specimens as units. The small change, however, is away from the present and dipole field directions, suggesting the removal of a small viscous component thus directed. Hence all specimens were heated to 350 °C and the directions after this heating have been used in the computation of the pole position, at 89° E, 57° N (B in Fig. 6). The pole positions obtained before treatment (A in Fig. 6), and after heating the pilot specimens to 450 °C (C), are also shown in Fig. 6. Differences between these positions are not sufficient to deviate the Manicouagan pole from the Triassic group, but it is possible that heating to 450 °C may give the best estimate of the dipole field direction at that time.

STABILITY TESTS

Viscous Magnetization (Thellier 1937)

Repeat measurements of natural remanent magnetization (n.r.m.) of 8 specimens stored in the earth's field for three months gave changes in excess

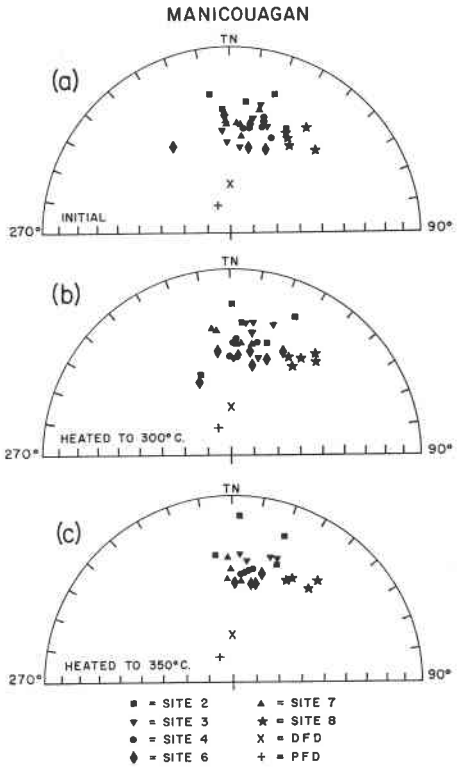


FIG. 2. Equal area stereograms showing: (a) initial specimen directions; (b) directions after heating to 300 °C; and (c) directions after heating to 350 °C. The present field direction (+), and dipole field direction (×), are also given.

of experimental error. The changes in intensity and direction are given in Table III. They are considered to be the result of a buildup of a temporary viscous magnetization (v.r.m.) along the direction of the field during storage; in most cases the viscous component was less than 5% of the n.r.m. component.

Four specimens, heated to 640 °C and cooled in zero field, were measured

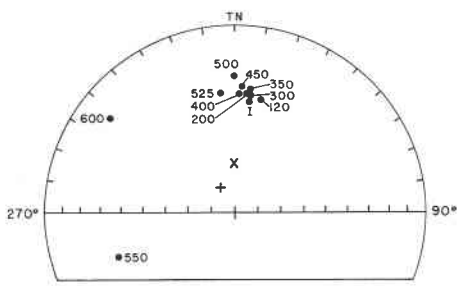


FIG. 3. Equal area stereogram showing mean direction of pilot group of 17 specimens after heating to successively higher temperatures, given in °C on diagram.

TABLE I

Site mean directions

Site	Initial						Heated to 300 °C						Heated to 350 °C					
	D	*I*	*N*	*R*	*k*	α	*D*	*I*	*N*	*R*	*k*	α	*D*	*I*	*N*	*R*	*k*	α
1	65	+41	5	2.913	2	78	55	+55	5	2.751	2	85						
2	8	+31	5	4.879	33	14	8	+32	5	4.778	18	19	9	+24	4	3.885	26	18
7	5	+41	6	5.965	144	6	1	+38	6	5.958	120	6	0	+40	4	3.985	199	7
6	5	+44	6	5.787	23	14	19	+46	6	5.860	36	11	10	+43	4	3.989	270	6
8	35	+39	5	4.971	139	7	37	+36	5	4.978	184	6	36	+35	4	3.983	179	7
3	8	+43	6	5.921	63	8	10	+35	6	5.926	68	8	12	+31	4	3.979	140	8
4	16	+41	5	4.977	181	6	5	+41	5	4.985	268	5	7	+41	4	3.998	1574	2
Mean of specimens	12	+41	33	32.093	35	4	11	+39	33	31.992	31	5	12	+36	24	23.410	39	5
Mean of sites	13	+40	6	5.920	63	9	12	+39	6	5.904	52	9	12	+36	6	5.897	49	10
9	310	−41	1															
10	300	+33	2				256	+45	2									

D is the declination measured clockwise from true north, *I* is the inclination, positive downwards, *N* is the number of specimens, *R* is the resultant giving each specimen unit weight, *k* is the best estimate of precision (Fisher 1953), and α is the half-angle of the cone of confidence about the mean at $P = 0.05$.

TABLE II

Pilot run formation means

Temp. °C	Sites as Units						Specimens as Units					
	D	*I*	*N*	*R*	*k*	α	*D*	*I*	*N*	*R*	*k*	α
20	12	+40	6	5.924	42	11	11	+40	17	16.670	39	6
120	11	+38	6	5.874	39	11	12	+38	12	11.831	65	5
200	11	+38	6	5.866	37	11	10	+39	12	11.810	58	6
300	13	+37	6	5.886	44	10	12	+37	17	16.633	44	6
350	12	+34	6	5.785	23	14	12	+36	12	11.715	39	7
400	5	+37	5	4.908	21	17	5	+36	12	11.645	24	9
450	8	+33	6	5.746	20	15	5	+33	13	12.354	19	10
500	5	+28	5	4.690	13	22	44	+27	12	11.093	12	13
525	359	+36	6	5.034	5	33	351	+29	15	11.170	4	23
550	353	+35	6	4.399	3	45	338	+33	15	9.539	3	30
600	322	+18	6	3.811	2	58	310	+5	15	8.551	2	35
650	266	+9	6	2.941	2	82	273	−8	15	7.988	2	38

NOTE: *D* is the declination measured clockwise from true north, *I* is the inclination, positive downwards, *N* is the number of sites or specimens, *R* is the resultant giving each site or specimen unit weight, *k* is the best estimate of precision (Fisher 1953), and α is the half-angle of the cone of confidence about the mean at a probability of 0.05.

TABLE III

Repeat measurement changes

Specimen	J_1/J_2	Angular Change
2B2*	.692	10°
3A1	.930	8°
4A	.950	14°
5A2	1.03	5°
6B2	1.02	9°
10A2	.973	5°
14B2†	1.48	28°
14C1†	.980	23°

NOTE: J_1/J_2 = ratio of intensity measurements; J_2 was made after the specimens had been stored in the earth's field for 3 months after measurement J_1.

*This specimen is from the site in which directions were random at P = 0.05 (Watson 1956).

†These specimens are from the site in which the grouping greatly improved after the specimens were heated to 300 °C.

after resting $\frac{1}{4}$ to $1\frac{1}{2}$ hours in the earth's field. They were measured again after 9 days in the earth's field. The intensities are given in Table IV. Assuming zero intensity at the time of removal from field free space, and a subsequent logarithmic buildup of viscous magnetization, the intensity of the v.r.m. at any time after the specimens have been subjected to a magnetic field may be calculated; values for $\frac{1}{2}$ and 1 hour are given in Table IV. Close agreement is evident between the experimental and theoretical values.

The large components (Table IV), ranging from 31 to 84% of the original n.r.m. intensity, built up in 9 days in thermally demagnetized specimens kept in the earth's magnetic field (56 000 γ), are in marked contrast to the small components (see Table III) built up under the same conditions in 3 months in untreated specimens. The latter are considered to be temporary components

TABLE IV
Viscous buildup

Specimen	J_1	$J_{\frac{1}{2}-1\frac{1}{2}}$	J_{214}	Theoretical $M_{\frac{1}{2}}$	Theoretical M_1
6A2	515	82	258	93	111
8B1	101	19	39	14	17
12B	574	257	481	173	208
14C2	298	43	93	33	40

NOTE: J_1 = intensity of n.r.m. in emu $\times 10^{-6}$; $J_{\frac{1}{2}-1\frac{1}{2}}$ = intensity measured $\frac{1}{2}$–$1\frac{1}{2}$ hours after removal from a null field oven in which specimen was heated to 640 °C, in emu $\times 10^{-6}$; J_{214} = intensity of same specimens after storage for 214 hours in the earth's magnetic field (56 000 γ); $M_{\frac{1}{2}}$ and M_1 = theoretical intensities in emu $\times 10^{-6}$ one half and one hour respectively after removal from null field.

(Creer 1957) that may be removed by thermal cleaning immediately prior to measurement. The former are thought to be v.r.m. components induced by the earth's magnetic field in magnetically 'soft' regions, which the specimens acquired by chemical or physical changes at a high (650 °C) temperature.

The precision peaks after heating to 350 °C for sites 3 and 6 (Fig. 2), which represent the fine and coarse material respectively, may be caused by the removal of a component thermally distributed in the 330–380 °C range. If the rock mass cooled slowly in the later stages, these lower temperature thermally distributed components might acquire directions corresponding to a considerable part of a secular variation cycle, so that they would tend to change site directions towards that of the dipole field direction. Removal of the low temperature distributed component by heating (i.e. to 350 °C, see Fig. 4) would then increase the between-site dispersion. These data are consistent with such a hypothesis but are inadequate to prove it. They do suggest, however, that highest precision in a group is not necessarily a criterion of greatest stability.

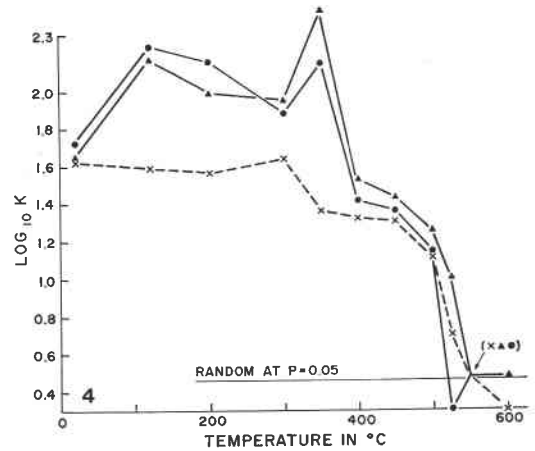
Representative thermal demagnetization curves from four sites (Fig. 5) show a small distributed component with a blocking temperature in the range 0–500 °C, and a large discrete component with a blocking temperature in the range 520–570 °C. This is consistent with the major component residing in minerals of nearly pure magnetite.

Polished Sections

Examination of a polished section from each of six sites revealed only minor variations in magnetic minerals between sites (Table V). Magnetite was the only magnetic mineral seen in significant amounts. The main variables were grain-size and the proportion of magnetite–ilmenite intergrowths.

CONCLUSION

The dacitic material from Manicouagan appears to contain a large n.r.m. component; viscous and other secondary components are small. The thickness of the formation, probably over 700 ft, and the spread of the site mean directions, suggest that most of the effect of the non-dipole field has been averaged



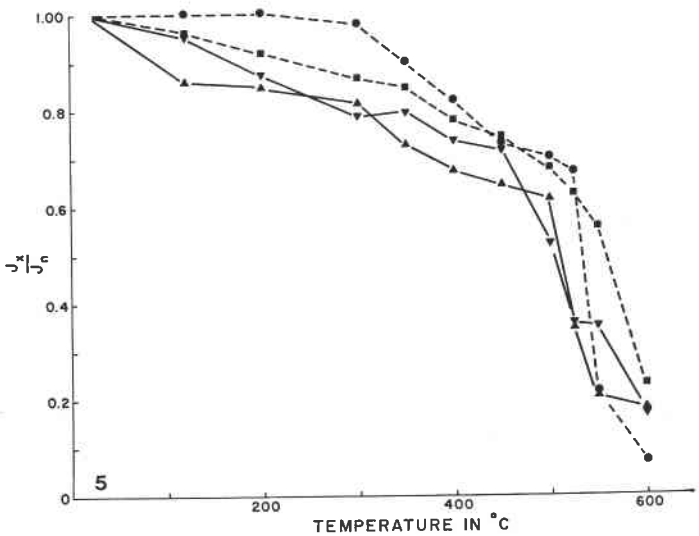


FIG. 4. Graph of $\log_{10} k$, the precision estimate of Fisher (1953), plotted against temperature in °C; ● = site 3, 4 specimens. ▲ = site 6, 4 specimens; ×- - -× ● site means, 6 sites.

FIG. 5. The mean intensity of magnetization for a site, after heating the specimens to the given temperature, divided by the initial intensity, (Jx/Jn), plotted against the heating temperature: ▲ = site 3, 4 specimens; ▼ = site 6, 4 specimens; ● = site 7, 2 specimens; ■ = site 8, 2 specimens.

out in the mean formation direction. The pole position from this direction, after thermal 'cleaning' at 350 °C, is 89° E, 57° N. This position (Fig. 6) lies in a group of Triassic poles from other North American formations, but is significantly different from the published North American poles of Permian, Jurassic, or Cretaceous age.

The mode of origin of the dacite is not clear from the magnetic data, but the pole position suggests a Triassic age.

TABLE V
Opaque Minerals

Site	Specimen	Size range	Magnetite	Ilmenite	Int.	Others	%	Remarks
1	2A	60–500 μ	P	D	P	—	3	Wide size range magnetite most common in smaller sizes
2	3A2	3–10 μ	D	P	—	Tr.Py.		Identification difficult on account of small grain size
3	12	50–150 μ	S	S				The rock contains ghosts of resorbed reconstituted crystals of large size (> 500 μ)
		10–50 μ				S.Py.		
		1–10 μ	D	P		S.Ch.		
6	8	60–250 μ	P	D	S	—	2–3	Many of the grains have sutured margins
7	5B	60-250 μ	P	P	P	—	3	Mainly anhedral grains with sutured margins.
8	10	60–300 μ	S	D	P	—	3	Most common opaques are ilmenite-ilmenite intergrowths.

NOTE: Int. = intergrowths of magnetite and ilmenite; % = approximate percentage of opaque minerals in the section; μ = Micron; D = dominant, P = prominant, S = subordinate, Tr = trace; Py. = Pyrite; Ch. = Chalcopyrite. No pyrrhotite was seen in polished section.

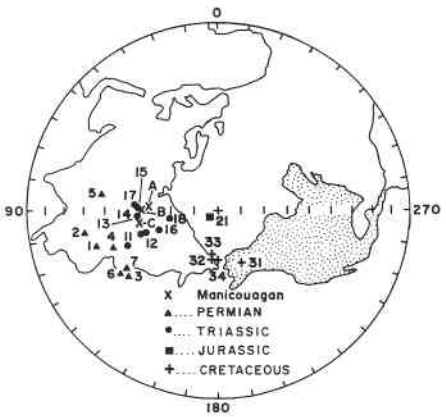


FIG. 6. Equal area polar projection of the northern hemisphere, showing the Manicouagan pole positions (A) before treatment, (B) after heating to 350 °C, (C) after heating to 450 °C. They are compared with Permian (▲), Triassic (●), Jurassic (■), and Cretaceous (+) poles from North American formations (Irving 1964). 1, Culter Formation (Graham 1955); 2, Abo Formation (Graham 1955); 3, Yeso Formation (Graham 1955); 4, Supai Formation (Collinson and Runcorn 1960); 5, Sangri de Cristo Formation (Graham 1955); 6, Prince Edward Island red beds (Roy 1963*a*); 7, Dunkard Series (Helsley 1965); 11, Chugwater Formation (Collinson and Runcorn 1960); 12, Moencopi Formation (Collinson and Runcorn 1960); 13, Springdale Sandstone (Runcorn 1956); 14, Chinle Formation (Collinson and Runcorn 1960); 15, Massachusetts lavas (Du Bois *et al.* 1957); 16, Newark Group (Opdyke 1961); 17, Connecticut Valley rocks (Du Bois *et al.* 1957); 21, White Mountain Series (Opdyke and Wensink 1966); 31, Monteregion Hills (Larochelle 1961); 32, Mount Megantic (Larochelle 1962); 33, Granite plutons, Sierra Nevada (Currie, Gromme, and Verhoogen 1963); 34, Isachsen Diabase (Larochelle and Black 1963).

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